

Paper 1

$$1. \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} v_0 \cos \theta t \\ v_0 \sin \theta t - \frac{1}{2} g t^2 \end{pmatrix}.$$

$$a) \quad h = v_0 \sin \theta t - \frac{1}{2} g t^2$$

$\frac{1}{2} g t^2 - v_0 \sin \theta t + h = 0$. By Vieta's theorem (hahaha):

$$t_1 + t_2 = \frac{2v_0 \sin \theta}{g}, t_1 t_2 = \frac{2h}{g}$$

$$t_1 - t_2 = \sqrt{(t_1 + t_2)^2 - 4t_1 t_2} = \frac{2\sqrt{v_0^2 \sin^2 \theta - 2gh}}{g}$$

$$D = v_0 \cos \theta (t_1 - t_2) = \frac{2v_0 \cos \theta \sqrt{v_0^2 \sin^2 \theta - 2gh}}{g}$$

$$b) \quad \text{Plug in } v_0 = 200, \theta = 45, h = 100, D = \mathbf{3872.31053 \text{ m}}.$$

Note: If you differentiate, you will find that maximum D is achieved at

$$\theta = \frac{\pi - \cos^{-1}\left(\frac{2gh}{v_0^2}\right)}{2}, \text{ but the question just wants maximum ground range.}$$

2.

$$a) \quad mg = kx, x = \frac{mg}{k} \text{ is the position the mass will come to rest.}$$

$$\text{Energy lost: } mgx - \frac{1}{2} kx^2 = \frac{m^2 g^2}{2k}$$

$$\text{Then } \Delta S = \frac{\Delta Q}{T} = \frac{m^2 g^2}{2kT} = 5.00979198 \times 10^{-4} \approx \mathbf{5.01 \times 10^{-4} \text{ J/K}}.$$

$$b) \quad i) \quad \sigma T^4 \cdot \frac{4\pi R_S^2}{4\pi R_{SE}^2} = I$$

$$T^4 = \frac{I \cdot R_{SE}^2}{\sigma \cdot R_S^2} = 1.06414109 \times 10^{15}$$

$$T = 5711.49503 \approx \mathbf{5710 \text{ K}}$$

$$ii) \quad \left(\sigma T_S^4 \cdot \frac{4\pi R_S^2}{4\pi R_{SM}^2} \right) \cdot \pi R_M^2 = \text{Power Received} = \sigma \cdot 4\pi R_M^2 \cdot T_M^4$$

$$\frac{T_S^4 R_S^2}{4R_{SM}^2} = T_M^4 = 2.47692887 \times 10^7$$

$$T_M = 223.089117 \approx \mathbf{223 \text{ K}}, \text{ which is surprisingly accurate.}$$

3. A derivation of the full SHM formulation is required.

a) $-kx = ma$

$a = -\frac{k}{m}x$, hence it oscillates at $\omega = \sqrt{\frac{k}{m}} = 10 \text{ rad/s}$

Let $x = A \cos(\omega t + \phi)$, $v = -A\omega \sin(\omega t + \phi)$, $a = -A\omega^2 \cos(\omega t + \phi)$

At $t = 0$, $x = A \cos \phi$, $v = -A\omega \sin \phi$. This leads to two results:

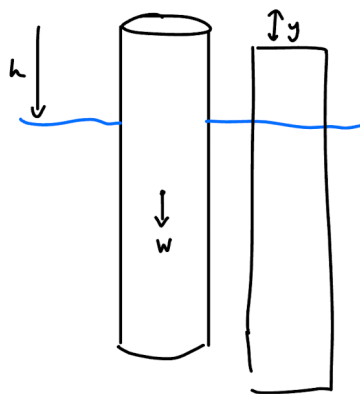
$$x^2 + \left(\frac{v}{\omega}\right)^2 = A^2 \rightarrow A = 0.1 \text{ m}$$

$$\tan \phi = \frac{-\frac{v}{\omega}}{x} = -\frac{v}{\omega x} \rightarrow \phi = -\frac{\pi}{3} \text{ rad}$$

$$v_{\max} = A\omega = 1 \text{ m/s}$$

$$a_{\max} = A\omega^2 = 10 \text{ m/s}^2$$

b)



push in by y , vertical position of CM: $-y$

Net force = Buoyancy - gravity

$$F = \Delta \text{Buoyancy}$$

$$m(-\ddot{y}) = \rho_w \Delta V g$$

$$= \rho_w A y g$$

$$\ddot{y} = -\frac{\rho_w A g}{m} y$$

Oscillation $\omega = \sqrt{\frac{\rho_w A g}{m}}$

we also know $\rho_w A(L-h)g = mg$

$$= \sqrt{\frac{g}{L-h}}$$

$$T = 2\pi \sqrt{\frac{L-h}{g}}$$

Extension: Cylinder on its side, which is not what the question is asking for

$$h = R(1 - \cos \theta) \rightarrow \frac{dh}{d\theta} = R \sin \theta \rightarrow \dot{h} = R \sin \theta \dot{\theta}$$

$\ddot{h} = R \sin \theta \ddot{\theta} + R \cos \theta \dot{\theta}^2$. Define $\theta = \theta_0 + \alpha$, small oscillation allows us to ignore all terms above 1st order: $\ddot{h} = R \sin \theta_0 \ddot{\alpha}$

$$A = \frac{1}{2}(2\pi - 2\theta)R^2 + \frac{1}{2}R^2 \sin 2\theta$$

$$\frac{dA}{d\theta} = -R^2 + R^2 \cos 2\theta = -R^2(1 - \cos 2\theta) = -2R^2 \sin^2 \theta$$

Using the same definition $\theta = \theta_0 + \alpha$, $\Delta A = -2R^2 \sin^2 \theta_0 \cdot \alpha$

$$ma = F_{net} = \rho \Delta V g = \rho l \Delta A g$$

$$m\ddot{h} = -2\rho R^2 l g \sin^2 \theta_0 \cdot \alpha$$

$$mR \sin \theta_0 \ddot{\alpha} = -2\rho R^2 l g \sin^2 \theta_0 \cdot \alpha$$

$$\ddot{\alpha} = -\frac{2\rho R l g \sin \theta_0}{m} \cdot \alpha$$

Then it oscillates with $\omega = \sqrt{\frac{2\rho R l g \sin \theta_0}{m}} = \sqrt{\frac{2\rho R l g \sqrt{1 - \left(1 - \frac{h}{R}\right)^2}}{m}} =$

$$\sqrt{\frac{2\rho R l g \sqrt{\frac{2h}{R} - \left(\frac{h}{R}\right)^2}}{m}} = \sqrt{\frac{2\rho l g \sqrt{2hR - h^2}}{m}}, \text{ hence } T = \frac{2\pi}{\omega} = \pi \sqrt{\frac{2m}{\rho l g \sqrt{2hR - h^2}}}$$

4.

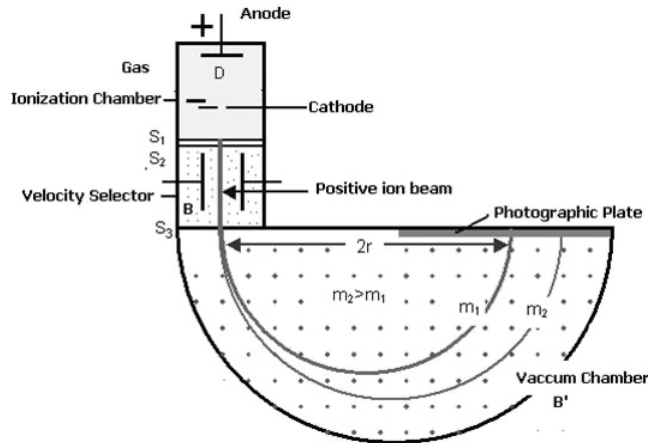
a) $V = \frac{kQ}{r}$. When the two drops combine, $Q_1 = 2Q$, $r_1^3 = 2r^3$

$$V_1 = \frac{kQ_1}{r_1} = \frac{2}{2^{1/3}}V = 2^{2/3} \times 1000 = 1587.40105 \approx \mathbf{1587V}$$

b) i) $q(\vec{E} + \vec{v} \times \vec{B}) = 0$, so $E = vB$. $v = \frac{E}{B} = \mathbf{5 \times 10^2 \text{ m/s}}$

ii) $qvB = \frac{mv^2}{R}$, $R = \frac{mv}{qB}$. He 2+

$$\Delta x = 2\Delta R = 2 \frac{\Delta mv}{qB} = 2.59375 \times 10^{-5} \approx 0.0259 \text{ mm} < 1 \text{ mm}, \text{ they cannot be resolved (i.e. they will go through the same slit).}$$



5.

a) Let N be the number of photons incident per area per unit time.

$$N \cdot \frac{hc}{\lambda} = I$$

$$P = \frac{F}{A} = \frac{dP}{dt} = N \cdot \left(2 \frac{h}{\lambda}\right) = \frac{2I}{C} = 3.33 \times 10^{-7} \text{ Pa}$$

b) $mR\omega^2 = \frac{kqq}{(2R)^2} \rightarrow R^3\omega^2 = \frac{kqq}{4m}$

We know that angular momentum $L = 2 \cdot mR^2\omega = n\hbar$ is quantized:

$$\frac{L^2}{4m^2R} = \frac{kqq}{4m}$$

$$\frac{L^2}{m \cdot kqq} = R$$

The energy state is $2 \times \frac{1}{2}mv^2 - \frac{kqq}{2R}$, and $\frac{mv^2}{R} = \frac{kqq}{(2R)^2}$

$$E = \frac{kqq}{4R} - \frac{kqq}{2R} = -\frac{kqq}{4R} = -\frac{m(kqq)^2}{4L^2} = -\frac{m\left(\frac{1}{4\pi\epsilon_0}qq\right)^2}{4n^2\left(\frac{h}{2\pi}\right)^2} = -\frac{mq^4}{16\epsilon_0^2h^2} \cdot \frac{1}{n^2}$$

$$\frac{hc}{\lambda_{min}} = \Delta E_{max} = \frac{mq^4}{16\epsilon_0^2h^2} \cdot \left(1 - \frac{1}{\infty^2}\right)$$

$$\lambda_{min} = \frac{16\epsilon_0^2h^3c}{mq^4} = \mathbf{183.514361 \text{ nm}}$$

Paper 2

$$1. \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} v_0 \cos \theta t \\ v_0 \sin \theta t - \frac{1}{2}gt^2 \end{pmatrix}.$$

$$a) \begin{pmatrix} v_0 \cos \theta t \\ v_0 \sin \theta t - \frac{1}{2}gt^2 \end{pmatrix} = \begin{pmatrix} d \\ -h \end{pmatrix}. \text{ Thus:}$$

$$t = \frac{d}{v_0 \cos \theta}$$

$$-h = d \tan \theta - \frac{gd^2}{2v_0^2} \sec^2 \theta$$

Now using the identity $\sec^2 \theta = 1 + \tan^2 \theta$:

$$-h = d \tan \theta - \frac{gd^2}{2v_0^2} (1 + \tan^2 \theta)$$

$$\frac{gd^2}{2v_0^2} (\tan^2 \theta) - d \tan \theta + \left(\frac{gd^2}{2v_0^2} - h \right) = 0$$

$$\tan \theta = \frac{d \pm \sqrt{d^2 - 4 \left(\frac{gd^2}{2v_0^2} \right) \left(\frac{gd^2}{2v_0^2} - h \right)}}{2 \cdot \frac{gd^2}{2v_0^2}}$$

$$\tan \theta = \frac{v_0^2}{gd} \left[1 \pm \sqrt{1 - 4 \left(\frac{g}{2v_0^2} \right) \left(\frac{gd^2}{2v_0^2} - h \right)} \right]$$

$$\tan \theta = \frac{v_0^2}{gd} \pm \sqrt{\frac{v_0^4}{g^2 d^2} + \frac{2v_0^2}{gd^2} h - 1}$$

$$\tan \theta = \frac{v_0^2}{gd} + \sqrt{\frac{v_0^4}{g^2 d^2} + \frac{2v_0^2}{gd^2} h - 1} = 1.38605555 \text{ or } 0.312891102$$

$$\theta = 54.1906140^\circ \text{ or } 17.3744372^\circ$$

$$b) \begin{pmatrix} v_0 \cos \theta t \\ v_0 \sin \theta t - \frac{1}{2}gt^2 \end{pmatrix} = \begin{pmatrix} d + v_1 t \\ -h \end{pmatrix}.$$

$$t = \frac{d}{v_0 \cos \theta - v_1}$$

$$-h = \frac{v_0 \sin \theta d}{v_0 \cos \theta - v_1} - \frac{1}{2}g \left(\frac{d}{v_0 \cos \theta - v_1} \right)^2$$

$$\frac{1}{2}g \left(\frac{d}{v_0 \cos \theta - v_1} \right)^2 - \frac{v_0 \sin \theta d}{v_0 \cos \theta - v_1} - h = 0$$

$$\frac{1}{2}gd^2 - v_0 \sin \theta d (v_0 \cos \theta - v_1) - h(v_0 \cos \theta - v_1)^2 = 0$$

$$\frac{1}{2}gd^2 - v_0^2 d \sin \theta \cos \theta + v_0 v_1 d \sin \theta - h(v_0^2 \cos^2 \theta - 2v_0 v_1 \cos \theta + v_1^2) = 0$$

$$-v_0^2(d \sin \theta \cos \theta + h \cos^2 \theta) + v_0(v_1 d \sin \theta + 2v_1 h \cos \theta) + \left(\frac{1}{2}gd^2 - hv_1^2 \right) = 0$$

$$v_0^2(d \sin \theta \cos \theta + h \cos^2 \theta) - v_0(v_1(d \sin \theta + 2h \cos \theta)) - \left(\frac{1}{2}gd^2 - hv_1^2 \right) = 0$$

$$v_0 = \frac{v_1(d \sin \theta + 2h \cos \theta) \pm \sqrt{v_1^2(d \sin \theta + 2h \cos \theta)^2 + 4(d \sin \theta \cos \theta + h \cos^2 \theta) \left(\frac{1}{2}gd^2 - hv_1^2 \right)}}{2(d \sin \theta \cos \theta + h \cos^2 \theta)}$$

SMH just ask the calculator to do it in the quadratic solver...

$$v_0 = \mathbf{60.6747615 \text{ or } 52.6032579 \text{ m/s}}$$

2. When a wave is reflected, it experiences π phase-shift. The distance difference must be a half-integer number of wavelengths.

- a) **Note: this question actually doesn't work, so I decided to change 90 degrees to 90.2, in order to make it actually in phase.**

$$x_1 = 120, x_2 = 2\sqrt{60^2 + 90.2^2}, x_3 = 2\sqrt{60^2 + (90.2 + h)^2}$$

$$x_3 - x_2 = \lambda$$

$$2\sqrt{60^2 + (90.2 + h)^2} - 2\sqrt{60^2 + 90.2^2} = \lambda$$

$$\sqrt{60^2 + (90.2 + h)^2} = \frac{\lambda}{2} + \sqrt{60^2 + 90.2^2}$$

$$60^2 + (90.2 + h)^2 = \left(\frac{\lambda}{2} + \sqrt{60^2 + 90.2^2} \right)^2$$

$$90.2 + h = \sqrt{\left(\frac{\lambda}{2} + \sqrt{60^2 + 90.2^2} \right)^2 - 60^2}$$

$$h = \sqrt{\left(\frac{\lambda}{2} + \sqrt{60^2 + 90.2^2} \right)^2 - 60^2} - 90.2 = 0.799606874 \approx \mathbf{0.800m}$$

b) $v = \sqrt{\frac{T}{\mu}}$ for speed of wave on a string. $v = f\lambda$

$$T = mg - \rho Vg \rightarrow T_1 = mg - \frac{\rho Vg}{2}, T_2 = mg - \rho Vg$$

For the second harmonic, $\lambda = L$

$$T_1 = mg - \frac{\rho Vg}{2} = \mu v_1^2 = \mu f_1^2 L^2$$

$$T_2 = mg - \rho Vg = \mu v_2^2 = \mu f_2^2 L^2$$

$$\frac{\rho Vg}{2} = T_1 - T_2 = \mu L^2 (f_1^2 - f_2^2)$$

$$\begin{aligned} \rho_{liquid} &= \frac{2\mu L^2}{\pi \left(\frac{d_{cyl}}{2}\right)^2 hg} (f_1^2 - f_2^2) = \frac{2\rho_w \pi \left(\frac{d_w}{2}\right)^2 L^2}{\pi \left(\frac{d_{cyl}}{2}\right)^2 hg} (f_1^2 - f_2^2) \\ &= \frac{2\rho_w d_w^2 L^2}{d_{cyl}^2 hg} (f_1^2 - f_2^2) = \mathbf{585.14377 \text{ kg/m}^3} \end{aligned}$$

Hence the mass can be found:

$$mg = 2T_1 - T_2 = \mu L^2 (2f_1^2 - f_2^2)$$

And the material's density:

$$\rho_{cyl} = \frac{\mu L^2 (2f_1^2 - f_2^2)}{Vg} = \frac{\rho_w d_w^2 L^2}{d_{cyl}^2 hg} (2f_1^2 - f_2^2) = \mathbf{5048.05713 \text{ kg/m}^3}$$

3.

- a) We have to find the expression for gravitational field, then integrate sequentially. Mass of the sphere: $M = \rho \cdot \frac{4}{3}\pi(R^3 - r^3)$

$$\text{For } x > R, g = -\frac{GM}{x^2}$$

$$\text{For } R \geq x > r, g = -\frac{G\rho\frac{4}{3}\pi(x^3 - r^3)}{x^2} = -\frac{4}{3}G\rho\pi\left(x - \frac{r^3}{x^2}\right)$$

$$\text{For } r \geq x, g = 0$$

After integration $\Delta V = -\int g \, dx$.

$$\text{For } x > R, V - 0 = -\int_{\infty}^x \left(-\frac{GM}{x^2}\right) dx = -\frac{GM}{x}$$

$$\text{For } R \geq x > r, -\frac{GM}{R} - V = -\int_R^x \frac{4}{3}G\rho\pi\left(x - \frac{r^3}{x^2}\right) dx$$

$$V = -\frac{GM}{R} + \frac{2G\rho\pi}{3}(x^2 - R^2) + \frac{4}{3}G\rho\pi r^3\left(\frac{1}{x} - \frac{1}{R}\right)$$

$$V = -\frac{\frac{4G\rho\pi}{3}(R^3 - r^3)}{R} + \frac{2G\rho\pi}{3}x^2 - \frac{2G\rho\pi R^2}{3} + \frac{\frac{4}{3}G\rho\pi r^3}{x} - \frac{\frac{4}{3}G\rho\pi r^3}{R}$$

$$V = -\frac{4G\rho\pi R^2}{3} + \frac{4G\rho\pi r^3}{3R} + \frac{2G\rho\pi}{3}x^2 - \frac{2G\rho\pi R^2}{3} + \frac{4G\rho\pi r^3}{3x} - \frac{4G\rho\pi r^3}{3R}$$

$$V = -\frac{6G\rho\pi R^2}{3} + \frac{2G\rho\pi}{3}x^2 + \frac{4G\rho\pi r^3}{3x}$$

$$V = -2G\rho\pi R^2 + \frac{2G\rho\pi}{3}x^2 + \frac{4G\rho\pi r^3}{3x}$$

$$\text{For } r \geq x, V = -2G\rho\pi R^2 + \frac{2G\rho\pi}{3}r^2 + \frac{4G\rho\pi r^3}{3r} = -2G\rho\pi R^2 + 2G\rho\pi r^2$$

$$\text{Ratio: } \frac{-\frac{GM}{R}}{-2G\rho\pi R^2 + 2G\rho\pi r^2} = \frac{\frac{\rho\frac{4}{3}\pi(R^3 - r^3)}{R}}{2\rho\pi(R^2 - r^2)} = \frac{2(R-r)(R^2 + Rr + r^2)}{3R(R-r)(R+r)} = \frac{2(R^2 + Rr + r^2)}{3R(R+r)}$$

Note: to check your answer, when $R = r$, $ratio = 1$; $r = 0$, $ratio = \frac{2}{3}$.

b) i) Heat flow: $\dot{Q} = \frac{kA\Delta T}{L} = \frac{kA(T_H - T_C)}{L}$

$$\text{Entropy change of hot / cold reservoir: } \dot{S}_H = -\frac{\dot{Q}}{T_H}, \dot{S}_C = +\frac{\dot{Q}}{T_C}$$

$$\begin{aligned} \dot{S}_{total} &= \dot{Q}\left(-\frac{1}{T_H} + \frac{1}{T_C}\right) = \frac{kA(T_H - T_C)}{L}\left(\frac{1}{T_C} - \frac{1}{T_H}\right) = \frac{kA(T_H - T_C)^2}{LT_H T_C} \\ &= \mathbf{0.129642845 \, J/Ks} \end{aligned}$$

SPhO 2019 Solution

ii) As we know, the angular momentum is quantized $L = n\hbar$, so this hydrogen is at a ground state. We are looking for the energies that can be used to excite it, which it will release later. The electron is non-relativistic. The highest energy transfer is a head-on, linear collision.

$$\begin{aligned} m_e v_0 &= m_e v_1 + m_h v_2 \\ \frac{1}{2} m_e v_0^2 &= \frac{1}{2} m_e v_1^2 + \frac{1}{2} m_h v_2^2 + E_{excite} \\ E_{excite} &= 13.6 \text{ eV} \left(1 - \frac{1}{n^2} \right) \end{aligned}$$

$$m_e (v_0^2 - v_1^2) = m_h v_2^2 + 2E_{excite}$$

Substitute out $v_1 = v_0 - \frac{m_h}{m_e} v_2$

Q3 b ii) $\left\{ \begin{aligned} m v_0 &= m v_1 + M v_2 \\ \frac{1}{2} m v_0^2 &= \frac{1}{2} m v_1^2 + \frac{1}{2} M v_2^2 + 2E_{excite} \end{aligned} \right.$

$$\begin{aligned} E_{excite} &= 13.6 \text{ eV} \left(1 - \frac{1}{n^2} \right) \\ &= \frac{hc}{\lambda} \end{aligned}$$

$$m v_0^2 = m \left(v_0 - \frac{M}{m} v_2 \right)^2 + M v_2^2 + 2E_x$$

$$v_0^2 = v_0^2 - \frac{2M}{m} v_0 v_2 + \frac{M^2}{m^2} v_2^2 + \frac{M}{m} v_2^2 + \frac{2E_x}{m}$$

$$0 = \left(\frac{M^2}{m^2} + \frac{M}{m} \right) v_2^2 - \left(\frac{2M}{m} v_0 \right) v_2 + \frac{2E_x}{m}$$

$$\Delta = \left(\frac{2M}{m} v_0 \right)^2 - 4 \left(\frac{M^2}{m^2} + \frac{M}{m} \right) \left(\frac{2E_x}{m} \right) \geq 0$$

$$\frac{\left(\frac{M}{m} v_0 \right)^2}{\left(\frac{M^2}{m^2} + \frac{M}{m} \right) \cdot \frac{2}{m}} \geq E_x$$

reduced mass

$$\frac{m M v_0^2}{2(M+m)} \geq E_x$$

$\rightarrow$ KE in the CM frame

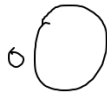
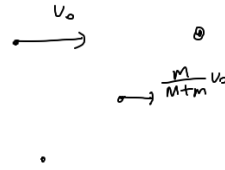
$$\frac{m}{13.6 \text{ eV}} \geq 1 - \frac{1}{n^2}$$

$$\eta \leq \sqrt{1 - \frac{m}{13.6 \text{ eV}}} = 3.246$$

$n = 2, 3, \dots$

Alternatively, look at the Zero Momentum Frame

CM frame (zero momentum frame)



Max energy you can suck out of system = KE in the ZMF

$$= \frac{1}{2} m \left(\frac{M}{M+m} v_0 \right)^2 + \frac{1}{2} M \left(\frac{m}{M+m} v_0 \right)^2$$

$$= \boxed{\frac{1}{2} \frac{Mm}{M+m} v_0^2}$$

$\equiv E_{\text{excite}}$

SPhO 2019 Solution

Substituting out v_2 is a slower solution:

$$\begin{aligned}
 m_e(v_0^2 - v_1^2) &= m_h \left(\frac{m_e(v_0 - v_1)}{m_h} \right)^2 + 2E_{excite} \\
 m_e(v_0^2 - v_1^2) &= \frac{m_e^2(v_0 - v_1)^2}{m_h} + 2E_{excite} \\
 (v_0^2 - v_1^2) &= \frac{m_e(v_0 - v_1)^2}{m_h} + \frac{2E_{excite}}{m_e} \\
 0 &= v_1^2 - v_0^2 + \frac{m_e}{m_h}(v_0^2 - 2v_0v_1 + v_1^2) + \frac{2E_{excite}}{m_e} \\
 0 &= v_1^2 \left(1 + \frac{m_e}{m_h} \right) - v_1 \left(2v_0 \frac{m_e}{m_h} \right) + \left[v_0^2 \left(\frac{m_e}{m_h} - 1 \right) + \frac{2E_{excite}}{m_e} \right]
 \end{aligned}$$

For this to have solution, discriminant $b^2 - 4ac \geq 0$

$$\begin{aligned}
 \left(2v_0 \frac{m_e}{m_h} \right)^2 - 4 \left(1 + \frac{m_e}{m_h} \right) \left[v_0^2 \left(\frac{m_e}{m_h} - 1 \right) + \frac{2E_{excite}}{m_e} \right] &\geq 0 \\
 \left(v_0 \frac{m_e}{m_h} \right)^2 &\geq \left(1 + \frac{m_e}{m_h} \right) \left[v_0^2 \left(\frac{m_e}{m_h} - 1 \right) + \frac{2E_{excite}}{m_e} \right] \\
 \frac{\left(v_0 \frac{m_e}{m_h} \right)^2}{\left(1 + \frac{m_e}{m_h} \right)} &\geq v_0^2 \left(\frac{m_e}{m_h} - 1 \right) + \frac{2E_{excite}}{m_e} \\
 \frac{\left(v_0 \frac{m_e}{m_h} \right)^2}{\left(1 + \frac{m_e}{m_h} \right)} + v_0^2 \left(1 - \frac{m_e}{m_h} \right) &\geq \frac{2E_{excite}}{m_e} = \frac{2}{m_e} \cdot 13.6eV \left(1 - \frac{1}{n^2} \right) \\
 \frac{m_e}{2 \times (13.6 eV)} \left[\frac{\left(v_0 \frac{m_e}{m_h} \right)^2}{\left(1 + \frac{m_e}{m_h} \right)} + v_0^2 \left(1 - \frac{m_e}{m_h} \right) \right] &\geq \left(1 - \frac{1}{n^2} \right) \\
 \frac{1}{n^2} &\geq 1 - \frac{m_e}{2 \times (13.6 eV)} \left[\frac{\left(v_0 \frac{m_e}{m_h} \right)^2}{\left(1 + \frac{m_e}{m_h} \right)} + v_0^2 \left(1 - \frac{m_e}{m_h} \right) \right] \\
 \sqrt{\frac{1}{1 - \frac{m_e}{2 \times (13.6 eV)} \left[\frac{\left(v_0 \frac{m_e}{m_h} \right)^2}{\left(1 + \frac{m_e}{m_h} \right)} + v_0^2 \left(1 - \frac{m_e}{m_h} \right) \right]}} &\geq n \\
 3.2468974 &\geq n
 \end{aligned}$$

Hence the energies $\frac{hc}{\lambda} = 13.6eV \left(1 - \frac{1}{n^2} \right)$

Wavelength: $\lambda = \frac{hc}{13.6eV \cdot \left(1 - \frac{1}{n^2} \right)}$

When $n = 2$, $\lambda_2 = 1.21875000 \times 10^{-7} \approx \mathbf{122nm}$

When $n = 3$, $\lambda_3 = 1.02832031 \times 10^{-7} \approx \mathbf{103nm}$

You're supposed to use the calculator instead of getting the expression first.

4. D

a) $F = qE = -\frac{Qq}{4\pi\epsilon_0} \frac{x}{(a^2+x^2)^{\frac{3}{2}}}$.

For very small x , $\frac{1}{(a^2+x^2)^{\frac{3}{2}}} = \frac{1}{a^3} \frac{1}{\left(1+\left(\frac{x}{a}\right)^2\right)^{\frac{3}{2}}} \approx \frac{1}{a^3} \left(1 - \frac{3}{2} \left(\frac{x}{a}\right)^2\right)$. However, we can just

ignore this as we are looking for first order terms only. Hence,

$$F = m\ddot{x} = -\frac{Qq}{4\pi\epsilon_0} \frac{x}{a^3}$$

$$\ddot{x} = -\frac{Qq}{4\pi\epsilon_0 m} \frac{x}{a^3}$$

It oscillates with $\omega = \sqrt{\frac{Qq}{4\pi\epsilon_0 ma^3}}$, $f = \frac{\omega}{2\pi} = \sqrt{\frac{Qq}{16\pi^3\epsilon_0 ma^3}}$

b) i) By Biot-savart's law, $dB = \frac{\mu_0 I}{4\pi} \frac{dl \times r}{r^3}$. Here we use the equivalence

$$I \times L = q \times v$$

Which is from $F = qvB = ILB$.

Then $B = \frac{\mu_0 q}{4\pi} \frac{v \times r}{r^3} = \frac{\mu_0 q}{4\pi} \frac{v}{r^2} = 3.93592026 \times 10^{-2} T$

ii) The charge of each ring $dq = \sigma \cdot 2\pi r dr$, and $v = r\omega = r \cdot 2\pi n$

$$\int_0^B dB = \int_0^R \frac{\mu_0 dq}{4\pi} \frac{v}{r^2}$$

$$B = \int_0^R \frac{\mu_0}{4\pi} \frac{v}{r^2} \sigma \cdot 2\pi r dr$$

$$B = \int_0^R \frac{\mu_0 \sigma r \cdot 2\pi n}{2} \frac{1}{r} dr$$

$$B = \int_0^R \pi \mu_0 \sigma n dr = \pi \mu_0 \sigma n R$$

5. Recall $\begin{pmatrix} x' \\ ict' \end{pmatrix} = \begin{pmatrix} \gamma & i\gamma\beta \\ -i\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} x \\ ict \end{pmatrix}$, or $\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}$.

a) i) We now have two events, $\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 5c \\ 4c \end{pmatrix}$

Transforming them into the other frame, we need $x' = 0$ for both. (Because the first event always gets 0.)

$$\begin{aligned} x' &= \gamma x - \gamma\beta ct \\ 0 &= \gamma \cdot 4c - \gamma\beta c \cdot 5 \\ 4\gamma c &= 5\gamma\beta c \\ \beta &= \frac{4}{5} \rightarrow \gamma = \frac{5}{3} \end{aligned}$$

ii) Now we transform for time:

$$\begin{aligned} ct' &= -\gamma\beta x + \gamma ct \\ ct' &= -\gamma\beta \cdot 4c + \gamma c \cdot 5 \\ t' &= \gamma(5 - 4\beta) = 3 \text{ s} \end{aligned}$$

b) Velocity addition formula: $u' = \frac{u-v}{1-\frac{uv}{c^2}}$ is the relative velocity (hence subtract)

$$\gamma' = \frac{1}{\sqrt{1 - \left(\frac{u'}{c}\right)^2}}$$

Length contraction: $l' = \frac{l}{\gamma'} = l \sqrt{1 - \left(\frac{u'}{c}\right)^2} = l \sqrt{1 - \left(\frac{\frac{u}{c} - \frac{v}{c}}{1 - \frac{uv}{c^2}}\right)^2}$

Transformation method (if you wanted to prove the formulas) : Back of cube

$B_1 = \begin{pmatrix} 0 \\ ct_1 \end{pmatrix}$, Front $F_1 = \begin{pmatrix} l \\ ct_1 \end{pmatrix}$. Same equations but using $-\beta$ in place of β , because we are going backwards.

$$x' = \gamma x + \gamma \beta ct$$

$$ct' = +\gamma \beta x + \gamma ct$$

In the earth frame: $B_0 = \begin{pmatrix} \gamma_1 \beta_1 ct_1 \\ \gamma_1 ct_1 \end{pmatrix}$, $F_0 = \begin{pmatrix} \gamma_1 l + \gamma_1 \beta_1 ct_1 \\ \gamma_1 \beta_1 l + \gamma_1 ct_1 \end{pmatrix}$

Now, going to the second frame from the zeroth frame:

$$x' = \gamma x - \gamma \beta ct$$

$$ct' = -\gamma \beta x + \gamma ct$$

$$B_2 = \begin{pmatrix} \gamma_2(\gamma_1 \beta_1 ct_1) - \gamma_2 \beta_2(\gamma_1 ct_1) \\ -\gamma_2 \beta_2(\gamma_1 \beta_1 ct_1) + \gamma_2(\gamma_1 ct_1) \end{pmatrix} = \gamma_1 \gamma_2 ct_1 \begin{pmatrix} \beta_1 - \beta_2 \\ 1 - \beta_1 \beta_2 \end{pmatrix}$$

$$F_2 = \begin{pmatrix} \gamma_2(\gamma_1 l + \gamma_1 \beta_1 ct_1) - \gamma_2 \beta_2(\gamma_1 \beta_1 l + \gamma_1 ct_1) \\ -\gamma_2 \beta_2(\gamma_1 l + \gamma_1 \beta_1 ct_1) + \gamma_2(\gamma_1 \beta_1 l + \gamma_1 ct_1) \end{pmatrix}$$

$$F_2 = \gamma_1 \gamma_2 \begin{pmatrix} l(1 - \beta_1 \beta_2) + ct_1(\beta_1 - \beta_2) \\ l(\beta_1 - \beta_2) + ct_1(1 - \beta_1 \beta_2) \end{pmatrix}$$

In order to measure distance, the time stamp must be the same:

$$\gamma_1 \gamma_2 ct_b(1 - \beta_1 \beta_2) = \gamma_1 \gamma_2 [l(\beta_1 - \beta_2) + ct_f(1 - \beta_1 \beta_2)]$$

$$ct_b = \frac{l(\beta_1 - \beta_2)}{(1 - \beta_1 \beta_2)} + ct_f$$

$$c(t_b - t_f) = \frac{l(\beta_1 - \beta_2)}{(1 - \beta_1 \beta_2)}$$

Now the distance difference:

$$l' = \gamma_1 \gamma_2 [l(1 - \beta_1 \beta_2) + ct_f(\beta_1 - \beta_2)] - \gamma_1 \gamma_2 ct_b(\beta_1 - \beta_2)$$

$$l' = \gamma_1 \gamma_2 [l(1 - \beta_1 \beta_2) + ct_f(\beta_1 - \beta_2) - ct_b(\beta_1 - \beta_2)]$$

$$l' = \gamma_1 \gamma_2 \left[l(1 - \beta_1 \beta_2) - \frac{l(\beta_1 - \beta_2)}{(1 - \beta_1 \beta_2)}(\beta_1 - \beta_2) \right]$$

$$l' = \frac{\gamma_1 \gamma_2 l}{(1 - \beta_1 \beta_2)} [(1 - \beta_1 \beta_2)^2 - (\beta_1 - \beta_2)^2]$$

$$l' = \frac{\gamma_1 \gamma_2 l}{(1 - \beta_1 \beta_2)} [\beta_1^2 \beta_2^2 + 1 - \beta_1^2 - \beta_2^2]$$

$$l' = \frac{\gamma_1 \gamma_2 l}{(1 - \beta_1 \beta_2)} (1 - \beta_1^2)(1 - \beta_2^2) = \frac{l}{\gamma_1 \gamma_2 (1 - \beta_1 \beta_2)}$$

If we simplify our previous answer:

$$\begin{aligned} l \sqrt{1 - \left(\frac{\frac{u}{c} - \frac{v}{c}}{1 - \frac{uv}{c^2}} \right)^2} &= l \sqrt{1 - \left(\frac{\beta_1 - \beta_2}{1 - \beta_1 \beta_2} \right)^2} = \frac{l}{(1 - \beta_1 \beta_2)} \sqrt{(1 - \beta_1 \beta_2)^2 - (\beta_1 - \beta_2)^2} \\ &= \frac{l}{(1 - \beta_1 \beta_2)} \sqrt{(1 - \beta_1^2)(1 - \beta_2^2)} = \frac{l}{\gamma_1 \gamma_2 (1 - \beta_1 \beta_2)} \end{aligned}$$

Satisfaction.